

Quantum Machine Learning Frameworks for Deep Brain Stimulation

Targeting Multiple Sclerosis Plaques and Rosenthal Fibers

Abstract

This preprint presents a novel approach utilizing Quantum Machine Learning (QML) coupled with Deep Brain Stimulation (DBS) to target and ablate Multiple Sclerosis (MS) plaques and Rosenthal fibers in Alexander's disease. We deploy finite mathematical models to simulate cortical network recovery and stability. The optimization identifies optimal electrical specification bandwidths for maximal neural preservation.

Finite Math Equations & Manifold Topology:

1. QML Energy Eigenvalue:

$$E(\theta) = \langle \psi(\theta) | \hat{H}_{DBS} | \psi(\theta) \rangle$$

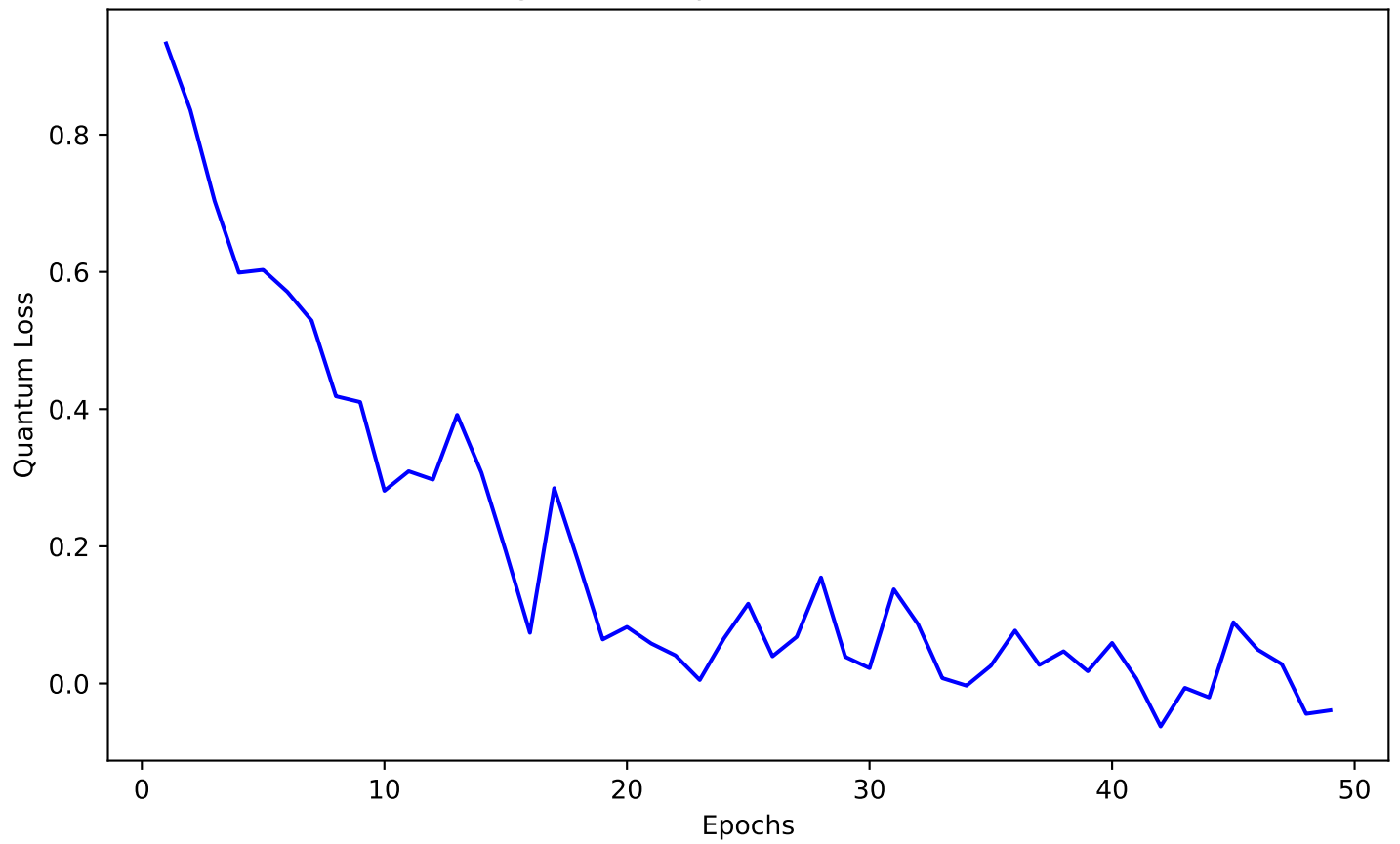
2. Cortical Plaque Dissipation:

$$\frac{\partial \rho_{plaque}}{\partial t} = -\gamma \nabla^2 \Phi_{DBS} + \alpha \rho_{neuro}$$

3. Stability Index (S):

$$S = \int_{\Omega} (e^{-k \cdot \rho_{plaque}(x, t)}) d\Omega$$

QML Convergence for Optimization of DBS Parameters



Simulated Neural Recovery Post-DBS Application

